

MAINTENANCE — THE BASE ON WHICH EVERYTHING IS ADDED

Definition: the maintenance requirement is the quantity of nutrients needed by an animal to keep **body weight, body composition and nitrogen balance constant**, with **no production and no work**, in a thermoneutral environment.

How it is measured — 3 methods: ① **Fasting catabolism / basal metabolism** — heat produced by a fasted, resting, thermoneutral animal (fasting heat production is the true zero). ② **Feeding trial / regression** — feed graded levels and find the intake at which weight change and N balance are zero. ③ **Comparative slaughter** — carcass energy compared against a baseline group.

Metabolic Body Size = $W^{0.75}$ (Kleiber's law – energy needs scale with surface area, not body weight)

Fasting Heat Production $\approx 70 \text{ kcal} \times W^{0.75}$ per day $\cdot$ ME for maintenance $\approx 125 \text{ kcal} \times W^{0.75}$

Why $W^{0.75}$ and not W: heat is lost through the body surface, and surface area is proportional to $W^{0.67-0.75}$. A 40 kg calf therefore needs far more energy **per kg body weight** than a 400 kg cow — the single most useful concept in this unit.

Practical ICAR-type maintenance figures for cattle: approximately **0.65 g DCP** and **7 g TDN** per kg body weight per day. So a **400 kg cow $\approx$ 260 g DCP + 2.8 kg TDN**; a 500 kg buffalo $\approx$ 325 g DCP + 3.5 kg TDN. **Field thumb rule:** ~1.5 kg concentrate for a cow, ~2 kg for a buffalo, plus roughage *ad libitum*.

Factors that raise the maintenance requirement: cold or heat stress, exercise and grazing distance (grazing adds **25–50%** over stall feeding), disease and parasitism, poor housing, high milk yield (larger visceral organs), age and breed (Bos indicus have a **lower** maintenance requirement than Bos taurus — a real advantage in India).

PRODUCTION REQUIREMENTS — QUICK REFERENCE

FUNCTION	REQUIREMENT OVER & ABOVE MAINTENANCE
Milk — cow (4% fat)	$\approx$ 45 g DCP + 300 g TDN per kg milk
Milk — buffalo (7% fat)	$\approx$ 60 g DCP + 400–450 g TDN per kg milk
Growth	Rises with ADG; protein need is highest per kg gain in the young (early gain is lean tissue & water; later gain is fat)
Pregnancy	60–70% of foetal growth occurs in the last 3 months $\Rightarrow$ allowance given only in the last trimester: extra 1–1.75 kg concentrate
Wool	Wool is pure keratin protein , rich in cystine (10–14% S-amino acids) $\Rightarrow$ requirement is for sulphur amino acids , not energy. 1 kg clean wool $\approx$ 1 kg protein deposited
Draft / work	Energy only (protein need barely rises). Light work +25% , medium +50% , heavy +75–100% over maintenance
Meat	Energy-dominated in the finishing phase; protein-dominated in the growing phase

The factorial principle — write this equation in any "requirement" answer:

Total requirement = Maintenance + Growth + Pregnancy + Lactation + Work

Each component is computed separately from the net requirement of the tissue, then divided by the **efficiency of utilisation** ($k_m \approx 0.72$ maintenance, $k_l \approx 0.62$ lactation, $k_f \approx 0.45$ fattening) to convert net to metabolisable requirement.

NUTRIENT PARTITIONING & PRIORITY — THE CONCEPT EXAMINERS LOVE

When nutrient supply is limited, the body does **not** share it equally. The **homeorhetic priority** order is:



The clinical consequence: reproduction is the **last** to be served and therefore the **first indicator of under-nutrition**. This is exactly why an under-fed herd shows **anoestrus, repeat breeding and long calving intervals** long before milk yield visibly drops. In early lactation the cow deliberately **mobilises body reserves** for milk even at the cost of fertility — the physiological basis of negative energy balance, ketosis and the post-partum anoestrus of Indian crossbred cows.

Body Condition Score (BCS, 1–5 scale): at calving **3.0–3.5**; at peak lactation not below **2.5**; at drying-off **3.0–3.5**. A loss of more than **1 BCS unit** in early lactation predicts ketosis and infertility. BCS is the cheapest field tool for judging whether your feeding standard is actually working.

GENERAL PRINCIPLES OF RATION COMPUTATION

Steps (write in order): ① Note the animal's **species, body weight, age, physiological state and level of production**. ② Read the **nutrient requirement** (DCP, TDN/ME, Ca, P, DM) from ICAR/NRC tables. ③ Estimate **DM intake capacity** (2.5–3% BW). ④ List **locally available feeds** with their composition and cost. ⑤ Fix the **roughage : concentrate ratio**. ⑥ Compute the concentrate mixture to fill the gap. ⑦ Add **mineral mixture 2% and salt 1%**. ⑧ **Check** total DM, NR, Ca:P and cost; adjust.

METHOD	HOW IT WORKS & WHEN TO USE
Pearson's Square	Balances one nutrient between two feeds/groups . Desired % in the centre, ingredient values at the corners; subtract diagonally and read parts. Simple, exam-friendly, but only one nutrient at a time
Trial and error	Ingredients are put together, totals computed, then adjusted repeatedly. Flexible, handles many nutrients; tedious
Algebraic / simultaneous equations	Two unknowns solved for two nutrient constraints. Precise for small mixtures
Linear programming (least-cost)	Computer optimisation with many ingredients and constraints. The industry method ; gives the cheapest ration meeting every constraint — but the answer is only as good as the constraints you set (must fix maximum inclusion limits, else it produces an unfeedable ration)
Thumb rules	Field shortcuts (1 kg concentrate per 2.5 L milk). Fast and practical; not precise

Pearson's Square worked in one line: to make an **18% CP** mixture from GNC (45% CP) and maize (9% CP) — differences: 18–9 = **9 parts GNC**; 45–18 = **27 parts maize**; total 36 $\Rightarrow$ GNC 25%, maize 75%.

THE DAIRY LIFE CYCLE — ONE CONTINUOUS FLOW



CALF FEEDING & EARLY WEANING

Colostrum — the non-negotiable numbers: feed within **30 minutes to 2 hours** of birth, at **10% of body weight per day** in 2–3 feeds, for the first **3–5 days**. Reason: the calf is born **agammaglobulinaemic** (epitheliochorial placenta blocks antibody transfer), and intestinal absorption of intact **IgG closes by 24–36 hours** ("gut closure"), with efficiency already halved by 12 h.

Colostrum vs normal milk: **higher** in immunoglobulins (IgG), total protein, vitamins A, D and E, minerals, and it is **laxative** (helps expel meconium); **lower** in lactose. It is roughly **5× the protein** and **10× the Vitamin A** of normal milk.

Milk feeding: whole milk at **1/10th of body weight** daily, gradually reduced. **Milk replacer** (CP 22–24%, fat 10–15%) can substitute after 3 weeks — must be milk-protein based, since the young calf cannot digest plant starch or non-milk protein.

Calf starter: CP **20–22%**, TDN 70–75%, palatable and highly digestible; offered from **2 weeks** of age.

Early weaning — the mechanism to write: the rumen develops in response to **VFA — especially butyrate and propionate — produced from fermenting the dry starter, not from milk** (milk bypasses the rumen through the oesophageal groove). Hence dry feed, not milk, grows the rumen papillae. Wean when the calf eats **1–1.5 kg of starter daily**, usually at **6–8 weeks**.

Advantages: saves saleable milk, cheaper rearing, earlier rumen development, less scours, easier management.

Calf scours prevention: clean utensils, fixed feeding time, milk at body temperature (~38 °C), never over-feed, never sudden change. Provide **clean water from week 1** — water is essential for rumen fermentation and cannot be replaced by milk.

HEIFER · PREGNANT · LACTATING · DRY · BULL

Growing heifer: aim for a steady **ADG of 400–500 g** (crossbred) so she attains **60–65% of adult body weight** at breeding and calves at **24–30 months**. **Do not over-feed energy** — excess fat is deposited in the udder as **fatty udder / mammary adipose infiltration**, permanently reducing lifetime milk yield. Under-feeding delays puberty and produces a small-framed, low-yielding cow.

Pregnant animal & "steaming up": in the **last 6–8 weeks** give an extra **1–1.75 kg** of concentrate to (i) meet rapid foetal growth (60–70% in the last trimester), (ii) build body reserves for the coming lactation, (iii) develop the udder, and (iv) improve calf birth weight and colostrum quality. **Avoid over-conditioning** — a fat cow at calving means dystocia, fatty liver, ketosis and retained placenta.

Challenge / lead feeding: begin about **2 weeks before calving**, increasing concentrate by **~0.5 kg per day** up to about **1% of body weight**, and continue increasing after calving as long as the cow responds with more milk. Purpose: adapt rumen papillae and microflora to the high-grain lactation diet **before** the demand arrives, and let the cow express her full peak. **Every extra 1 kg at peak yield ≈ 200 kg extra milk over the lactation**.

Lactating cow: peak yield at **6–8 weeks** but peak DM intake only at **10–12 weeks** — this lag creates the inevitable **negative energy balance** of early lactation. Feed high-energy-density rations, bypass fat, and 4–6 small concentrate meals rather than two large ones.

Dry period — 60 days: allows udder tissue regeneration, replenishment of body reserves and foetal growth. A dry period shorter than 45 days cuts the next lactation by 20–25%. Feed mainly roughage; keep **Ca low** in the last 2–3 weeks to prime the milk-fever defence.

Breeding bull: maintenance ration plus **1–1.5 kg extra concentrate**; keep him in **breeding, not fattening, condition**. Ensure Vitamin A, Zn, Se and adequate exercise — obesity causes low libido, poor sperm motility and leg problems.

Working bullock: energy is the limiting nutrient. Feed the extra ration **after work, not before**; give ample water and salt; add **25–100% over maintenance** depending on work intensity.

SHEEP & GOAT FEEDING

POINT	SHEEP	GOAT
Feeding habit	Grazer — bites close to the ground, selective for leaf	Browser — prefers tree leaves & shrubs at height; highly selective
DM intake	3–4% BW	3–5% BW
Tolerance	Very sensitive to copper toxicity	Tolerates tannins & bitter browse better than any other ruminant
Product focus	Meat + wool (S-amino acids)	Meat + milk (highest yield per kg body weight)

Phase feeding of the ewe/doe — the four terms you must define:

- **Flushing** — extra **250–500 g** of concentrate (or improved grazing) beginning **2–3 weeks before** mating and continuing 2–3 weeks into the breeding season. Purpose: raise body condition and **ovulation rate** ⇒ more twins and a tighter lambing.
- **Steaming up** — extra concentrate in the **last 6 weeks** of gestation, when 70% of foetal growth occurs; prevents **pregnancy toxæmia** and gives heavier, more viable lambs with better colostrum.
- **Creep feeding** — a palatable **18–20% CP** creep ration offered to lambs/kids from **2–3 weeks** in an enclosure the dam cannot enter; speeds rumen development and weaning weight.
- **Fattening / finishing** — high-energy ration (concentrate 60–70%) for the last 60–90 days before slaughter.

Indian field realities: pregnancy toxæmia in twin-bearing **Marwari and Muzaffarnagri** ewes on poor late-gestation grazing; **urolithiasis** in male sheep and goats on high-concentrate, high-P, low-water diets — prevent with **Ca:P 2:1, 1–2% ammonium chloride, up to 4% salt** and abundant water; and **copper toxicity** when sheep are fed cattle mineral mixture (never interchange them).

BYPASS NUTRIENTS — THE HIGH-YIELDER TECHNOLOGY

The problem: in a high-yielding cow, rumen fermentation destroys exactly what she needs most. Good-quality dietary protein is degraded to ammonia and rebuilt into microbial protein of **fixed, average quality**, while dietary fat is **biohydrogenated** and, above 3–4% of DM, poisons the fibre-digesting bacteria. The animal's genetic potential exceeds what rumen fermentation alone can supply.

BYPASS (RUMEN-PROTECTED) PROTEIN — definition: dietary protein that **escapes rumen degradation** and is digested and absorbed as amino acids in the small intestine (= RUP or UDP).

How it is made: **chemical** — formaldehyde treatment at ~1 g HCHO per 100 g of **crude protein** (0.8–1.2%), forming reversible methylene bridges that are stable at rumen pH 6.5 and hydrolysed in the acidic abomasum; also tannic acid. **Physical** — controlled heating, roasting, expeller/extrusion. **Natural** — feeds inherently high in RUP: fish meal, maize gluten, distillers' grains.

Typical Indian practice: formaldehyde-treated **mustard cake or groundnut cake** fed to crossbred cows and Murrah buffaloes. **Benefits:** milk yield up **10–15%**, higher milk protein, improved N efficiency, less urinary N loss, better body condition and earlier resumption of cyclicity.

BYPASS FAT: **calcium salts of long-chain fatty acids (Ca-LCFA)** or hydrogenated **prilled fat** — inert at rumen pH, dissociating in the abomasum. Fed at **100–150 g/day**. Raises the energy density of the ration **without depressing fibre digestion or causing acidosis** — the single best tool against early-lactation negative energy balance.

The balance to state: optimum **RDP : RUP is about 65 : 35** for high yielders. Excess bypass protein is wasteful and starves the microbes of ammonia; too little leaves the cow short of intestinal amino acids. **Bypass protein is useless unless the rumen is first given enough RDP and fermentable energy.**

NON-PROTEIN NITROGEN (NPN) FOR RUMINANTS — USE, LIMITS AND TOXICITY

POINT	DETAIL TO WRITE
What is NPN	Nitrogenous compounds that are not true protein: urea , biuret, ammonium salts, amides, amines, nitrates, and the natural NPN of green fodder and silage
Why it works only in ruminants	Rumen bacteria produce urease , splitting urea into ammonia + CO₂ ; the NH ₃ is then combined with carbon skeletons and ATP from fermentable energy to form microbial crude protein . Non-ruminants have no ureolytic microbes ahead of the site of absorption, so urea is simply toxic to them
Protein equivalent	Urea has 45–46.6% N ⇒ × 6.25 = 281–291% CP equivalent . 1 kg urea ≈ the N of ~2.8 kg of groundnut cake
Safe limits — memorise	Maximum 1% of the concentrate mixture, 3% of total DM , and never more than one-third of the total dietary N . Adapt gradually over 2–3 weeks
Essential co-factors	A readily fermentable energy source (molasses, grain starch) must be present at the same time; N : S ratio 10–14 : 1 ; adequate Co, P and trace minerals
Absolute prohibitions	Never in drinking water (rapid absorption); never to young pre-ruminant calves , non-ruminants, sick or starved animals; never mixed unevenly; never with raw soybean (its urease releases ammonia instantly)
UMMB — Urea Molasses Mineral Block	Typical composition: molasses 40–50% , urea 10% , oil cake 15–20% , mineral mixture 10% , salt 5% , binder (cement/bentonite/quicklime) 10% . A licking block that delivers N, energy and minerals slowly and safely to grazing animals on dry-season straw — the flagship Indian NPN technology

UREA (AMMONIA) TOXICITY — the full answer. **Cause:** overdose, sudden introduction, uneven mixing, or feeding without fermentable energy ⇒ NH₃ released faster than the microbes can use it. **Mechanism:** rumen pH rises above **7.5–8.0**; ammonia now exists in the un-ionised (NH₃) form, which is **freely absorbed** across the rumen wall; blood ammonia exceeds the liver's urea-cycle capacity (> ~1 mg/dL) and is neurotoxic. **Signs (onset 20 min – 4 h):** uneasiness, muscle tremors, salivation, bloat, grinding of teeth, incoordination, laboured breathing, violent struggling, tetany, and death. **Treatment:** drench **4–5 litres of 5% acetic acid (vinegar)** to acidify the rumen and trap ammonia as the non-absorbable NH₄⁺ ion, plus **large volumes of cold water** to dilute and slow fermentation; repeat as needed; trocarisation for bloat. **Lethal dose ≈ 0.3–0.5 g urea per kg body weight.**

THE SEVEN CLASSICAL PRODUCTION DISEASES — ONE ROW EACH

DISORDER	WHEN & WHO	CAUSE / MECHANISM	SIGNS & LAB FINDING	TREATMENT	NUTRITIONAL PREVENTION
Milk Fever (Parturient paresis)	Within 72 h of calving ; high yielders in 3rd–6th lactation ; Jersey & crossbreds	Sudden massive Ca drain into colostrum; the Ca-mobilising machinery (PTH, 1,25-(OH) ₂ -D ₃ , bone resorption) has not yet been activated	Three stages: excitement/tremor → sternal recumbency with S-shaped neck kink → lateral recumbency & coma. Serum Ca falls below 5–6 mg/dL (normal 9–12); subnormal temperature, dry muzzle, no rumen motility	300–500 mL of 25% calcium borogluconate slowly IV (monitor heart), part SC; oral Ca gels	Low-calcium diet in the last 2–3 weeks so the Ca-mobilising system stays switched on; negative DCAD of –50 to –100 mEq/kg DM using anionic salts; adequate Mg; Vitamin D ₃
Ketosis (Acetonaemia)	3–6 weeks after calving — peak yield, peak negative energy balance	Energy demand exceeds intake ⇒ massive fat mobilisation ⇒ acetyl-CoA exceeds the oxaloacetate available (oxaloacetate is drained for gluconeogenesis & lactose) ⇒ ketone bodies accumulate	Drop in appetite (refuses concentrate first) and milk yield, rapid weight loss, sweetish acetone smell of breath, milk and urine , constipation, nervous form (licking, circling). Blood glucose < 40 mg/dL ; ketones ↑ in urine/milk (Rothera’s test)	500 mL of 25% dextrose IV ; propylene glycol 200–300 mL orally twice daily (glucogenic); glucocorticoids; sodium propionate; cobalt & B ₁₂	Avoid over-fat cows at calving (BCS ≤3.5); challenge/lead feeding ; energy-dense early-lactation ration; bypass fat ; niacin, propylene glycol, monensin
Ruminal Acidosis (Grain overload; SARA in its subacute form)	Sudden access to grain; challenge feeding; high-concentrate feedlot & high-yielder diets	Rapid fermentation of starch ⇒ Streptococcus bovis then Lactobacillus proliferate ⇒ lactic acid accumulates ⇒ rumen pH crashes; cellulolytic bacteria and protozoa die; osmotic pull of fluid into the rumen ⇒ dehydration; systemic metabolic acidosis	Acute: rumen pH < 5.0 , rumen stasis, profuse pasty diarrhoea, dehydration, staggering, recumbency, death. SARA: pH 5.2–5.6 intermittently — milk-fat depression, erratic intake, loose faeces with undigested grain, laminitis and liver abscesses	Oral alkalinising agents (magnesium hydroxide/sodium bicarbonate), IV fluids with bicarbonate, thiamine, rumen lavage or rumenotomy in severe cases; transfaunation	Gradual adaptation to grain ; keep effective fibre — minimum 18–21% ADF / ~28% NDF , with adequate particle length; sodium bicarbonate 0.75–1.5% of DM ; TMR feeding to stop slug-feeding; ionophores
Bloat (Ruminal tympany)	Frothy — lush legume grazing (lucerne, berseem) or finely ground grain. Free-gas — obstruction, oesophageal choke, vagal indigestion	Frothy: soluble leaf proteins and saponins stabilise a stable foam , so gas cannot be eructated. Free-gas: physical failure of eructation	Distension of the left paralumbar fossa , dyspnoea, mouth breathing, tongue out, death from asphyxia and cardiac compression	Anti-foaming agents — poloxalene , vegetable/mineral oil 250–500 mL, turpentine; stomach tube for free gas; trocar and cannula in emergency	Never turn hungry animals onto lush legume pasture; feed dry straw before grazing ; keep legume below half of the pasture; wilt the fodder; poloxalene or ionophores in feed
Hypomagnesaemic tetany (Grass tetany / Grass staggers)	Lactating cows on lush young spring grass heavily fertilised with N and K; milk tetany in milk-fed calves	Low Mg intake plus high K and N in young grass blocking Mg absorption from the rumen; no meaningful body store of Mg is available	Hyperexcitability, muscle fasciculation, staggering, tetanic convulsions with opisthotonus , sudden death. Serum Mg < 1.7 mg/dL	Magnesium sulphate solution IV/SC (usually combined with calcium borogluconate)	Mg supplementation (MgO in concentrate or Mg bullets) during the risk period; feed hay before turning out; avoid heavy K fertilisation
Pregnancy Toxaemia (Ovine ketosis / twin-lamb disease)	Last 4–6 weeks of gestation in twin- or triplet-bearing ewes and does	Multiple foetuses demand glucose while the enlarged uterus physically restricts rumen capacity ⇒ severe negative energy balance ⇒ ketosis and hypoglycaemia	Separation from the flock, blindness, star-gazing, teeth grinding, incoordination, recumbency, ketonuria, death within 2–7 days	IV glucose, oral propylene glycol/glycerol, corticosteroids, and induction of parturition or caesarean as the definitive cure	Steaming up in the last 6 weeks; avoid both over-fat and thin ewes at mating; never starve or transport in late gestation; small frequent energy-dense meals
Post-parturient Haemoglobinuria	High-yielding buffaloes , within weeks of calving — a classic Indian condition	Hypophosphataemia , often with heavy berseem feeding and cruciferous fodders; low ATP in erythrocytes ⇒ intravascular haemolysis	Sudden coffee-coloured urine , pallor, jaundice, weakness, drop in milk. Serum P falls below 2 mg/dL	Sodium acid phosphate 60 g in 300 mL water IV , repeated; blood transfusion in severe cases	Adequate dietary phosphorus and mineral mixture; limit berseem to part of the roughage; maintain Ca:P near 2:1

THE UNIFYING PRINCIPLE — WRITE THIS AS YOUR CONCLUSION

Every production disease above is the same story: **the modern high-yielding animal's output demand at a single moment exceeds the rate at which her homeostatic machinery can supply the nutrient** — calcium at calving, glucose at peak lactation, buffering capacity at grain overload, magnesium on lush pasture, phosphorus in the freshly calved buffalo. They are therefore **diseases of production, not of infection**, they cluster around **the transition period (3 weeks before to 3 weeks after calving)**, and they are all **prevented by feeding management rather than cured by drugs**. The three interventions that prevent most of them are: **correct dry-period feeding (BCS 3.0–3.5, low Ca, negative DCAD)**, **gradual dietary transition with challenge feeding**, and **adequate effective fibre with buffers**.

FEEDING DURING DROUGHT, FLOOD & SCARCITY

Principle: in scarcity the objective changes from **production to survival** — keep the breeding nucleus alive and pregnant, accept the loss of milk yield.

Strategies to list: ① **Survival ration** — as low as 50–60% of maintenance for short periods; feed at least **1% of body weight as roughage** to keep the rumen functional. ② **Urea-ammoniated straw** (4 kg urea + 40 L water per 100 kg straw, 21 days). ③ **UMMB licks** for grazing animals. ④ **Unconventional and agro-industrial by-products** — sugarcane bagasse and tops, tree leaves (subabul, neem, babul, pipal), cactus, seaweed, molasses, deoiled cakes, vegetable and fruit waste, poultry droppings (where permitted). ⑤ **Complete feed blocks / densified total mixed ration blocks** — easy to transport to a scarcity zone. ⑥ **Culling** unproductive stock to save feed for the productive core. ⑦ Fodder banks, hay and silage reserves made in the flush season. ⑧ Ensure clean water and mineral mixture — deficiency kills faster than hunger.

Flood-specific: feeds get wet and mouldy ⇒ **aflatoxin risk**; provide dry, elevated storage, toxin binders, and watch for nitrate poisoning from flooded, heavily manured forage.

READY-MADE RATION TEMPLATES

ANIMAL	A DEFENSIBLE DAILY RATION
400 kg cow, 10 L milk (4% fat)	Green fodder 25 kg + wheat straw 4 kg + concentrate mixture (CP 18%, TDN 70%) 5.5 kg (1.5 maintenance + 4 for milk) + mineral mixture 50 g + salt 30 g
500 kg buffalo, 8 L milk (7% fat)	Green fodder 25 kg + paddy straw 5 kg + concentrate 6 kg (2 maintenance + 4 for milk) + bypass fat 100 g + mineral mixture 60 g + salt 35 g
Growing calf, 100 kg	Green fodder 8–10 kg + calf starter/concentrate (CP 20%) 1.5 kg + mineral mixture 20 g + clean water
Adult ewe / doe, 30 kg	Grazing 6–8 h + green fodder 3 kg + concentrate (CP 14–16%) 200–300 g ; double the concentrate during flushing & late gestation
Working bullock, 400 kg (medium work)	Maintenance ration + 50% extra energy : green fodder 20 kg + straw 5 kg + concentrate 2.5–3 kg , fed after the day's work, with ample water and salt

Always close a ration answer with the four checks: total DM within 2.5–3% of body weight · nutritive ratio matched to the stage · Ca:P near 2:1 · mineral mixture and salt included.

LAST-10-MINUTES DRILL — UNIT 3

Metabolic body size — $W^{0.75}$ (Kleiber)
Fasting heat production — $\sim 70 \text{ kcal} \times W^{0.75}$
Maintenance, cattle — $\sim 0.65 \text{ g DCP} + 7 \text{ g TDN per kg BW}$
Per kg cow milk (4% fat) — $45 \text{ g DCP} + 300 \text{ g TDN}$
Per kg buffalo milk (7% fat) — $60 \text{ g DCP} + 400\text{--}450 \text{ g TDN}$
Foetal growth in last trimester — 60–70%
Wool requirement is for — Sulphur amino acids (cystine)
Heavy work energy increase — +75–100% over maintenance
Colostrum timing & amount — Within 2 h, 10% of body weight/day
Gut closure to IgG — 24–36 hours
Calf starter CP — 20–22%
Milk fed to calf — 1/10th of body weight
What develops the rumen — VFA (butyrate, propionate) from dry starter
Early weaning age — 6–8 weeks (1–1.5 kg starter/day)
Heifer breeding weight — 60–65% of adult body weight
Danger of over-feeding heifers — Fatty udder ⇒ lifelong low yield
Dry period length — 60 days
Steaming up — Extra 1–1.75 kg concentrate, last 6–8 weeks
Challenge feeding start & rate — 2 wk pre-calving, +0.5 kg/day
Peak yield vs peak intake — 6–8 wk vs 10–12 wk (⇒ NEB)
1 kg extra at peak = — $\sim 200 \text{ kg extra per lactation}$
BCS at calving — 3.0–3.5 (of 5)
Flushing — +250–500 g, 2–3 wk before mating ⇒ ↑ ovulation
Creep feed CP — 18–20%
Grazer vs browser — Sheep vs goat
Species most sensitive to Cu — Sheep
Formaldehyde for bypass protein — $\sim 1 \text{ g per } 100 \text{ g CP}$
Bypass fat form & dose — Ca salts of LCFA, 100–150 g/day
Ideal RDP : RUP — 65 : 35
Urea max in concentrate / total DM — 1% / 3%
Urea protein equivalent — 281–291%
Urea toxicity antidote — 4–5 L of 5% acetic acid + cold water
Urea lethal dose — 0.3–0.5 g/kg body weight
UMMB urea & molasses % — 10% urea, 40–50% molasses
Milk fever serum Ca — < 5–6 mg/dL
Milk fever treatment — 300–500 mL 25% Ca borogluconate IV
DCAD formula & target — $(\text{Na}+\text{K})-(\text{Cl}+\text{S})$; –50 to –100 mEq/kg DM
Ketosis blood glucose — < 40 mg/dL
Ketosis oral glucogenic — Propylene glycol 200–300 mL BID
Acute acidosis rumen pH — < 5.0 (SARA 5.2–5.6)
Bacteria of grain overload — Strep. bovis → Lactobacillus
Minimum effective fibre — 18–21% ADF / ~28% NDF
Buffer dose — NaHCO_3 0.75–1.5% of DM
Frothy bloat treatment — Poloxalene / oil; trocar in emergency
Grass tetany serum Mg — < 1.7 mg/dL
Pregnancy toxemia — who — Twin-bearing ewes, last 4–6 weeks
Coffee-coloured urine in buffalo — Post-parturient haemoglobinuria (low P)
Its treatment — Sodium acid phosphate IV
Urolithiasis prevention — Ca:P 2:1, NH_4Cl 1–2%, salt 4%, water
Least-cost formulation method — Linear programming
One-nutrient two-feed method — Pearson's square
Survival ration roughage minimum — 1% of body weight